

Comprehensive Digital Forensic Investigation Report: Addressing a Data Breach at Dark Clarke

(COMP5004)

Submitted by:

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Introduction:

Cybersecurity incidents are becoming more common and pose a danger to firms in a variety of industries in the current digital age. A data breach or cyberattack can have disastrous repercussions, from monetary losses and harm to one's reputation to legal troubles and fines. Therefore, it is impossible to overestimate the significance of digital forensic investigations, especially in the cybersecurity sector.

Digital forensics is a specialised field that identifies, gathers, examines, and analyses digital evidence using computer science, legal principles, and investigation tactics. It is essential to incident response and breach investigations because it helps companies identify the cause of an issue, gauge its extent and consequences, and obtain data that may be utilised in court or for regulatory compliance.

Digital forensic investigations are particularly important to the cybersecurity business. The sensitive data and systems of their clients are entrusted to cybersecurity firms like DarkClarke. These businesses need to respond quickly and forcefully in the event of a security issue, like a data leak, to locate the breach's origin, minimise damage, and put the necessary corrective actions in place.

The intricate and demanding nature of digital forensic investigations within the cybersecurity sector is a result of the sophisticated tactics used by threat actors and the sophisticated nature of cyber-attacks. Cybercriminals may employ diverse strategies for concealing their actions and avoiding discovery, such as obfuscation techniques, anti-forensic procedures, and data-hiding techniques. Therefore, to properly analyse and respond to incidents, cybersecurity experts need to have a thorough understanding of digital forensics and use state-of-the-art tools and procedures.

The main steps of the digital forensic investigation process that I, an impartial digital forensic expert, would conduct to help Dark Clarke deal with the data breach problem are described in this report. Because of Dark Clarke's experience with cybersecurity and the possibility that the perpetrator(s) used sophisticated data-hiding methods, this investigation needs to be conducted thoroughly and methodically, utilizing cutting-edge forensic tools and procedures.

Dark Clarke can determine the origin of the data breach, comprehend the techniques employed by the offender(s), and put in place the necessary steps to lessen the incident's effects, bolster its security posture, and stop it from happening again by carrying out a comprehensive digital forensic investigation. Furthermore, the investigation's conclusions can offer insightful analysis and important takeaways for the larger cybersecurity community, bolstering the group's efforts to fend off cyberattacks and safeguard private information.

Protecting the integrity and admissibility of evidence involves several essential steps in the digital forensics' investigation process. These procedures include

1. Identification phase
2. Preparation phase
3. Acquisition phase
4. Preservation phase
5. Examination and analysis phase

6. Reporting and presentation phase

These procedures are intended to carefully collect, store, examine, and display digital evidence in a way that adheres to digital forensics standards.

2. Acquisition:

While investigating an occurrence, a digital forensics investigator may want to ask a manager certain important questions before starting the investigation such as

- What is the suspected incident/breach's nature and extent?
- What are the functions of the important individuals and users of interest?
- Which vital applications, systems, and data assets are at stake?
- What logging, security measures, and recent modifications are in place in your environment?
- What actions have been taken in response to incidents thus far?
- What operational restrictions and priorities do we need to consider?
- Who will oversee establishing the investigation's main point of contact?

The objective is to swiftly assimilate event context, pinpoint important data sources and personnel, comprehend the IT environment and security posture of the organization, coordinate protocols and priorities, and set up channels of communication and coordination for the investigation.

2.1. Identification phase:

2.1.1. Identifying potential data sources:

Finding any probable data sources that might have pertinent information concerning the data leak will be the first step in the inquiry. Among these data sources are, but are not restricted to: (ICIJ, n.d.) (Ekran, n.d.)

- Network traffic logs: Analysing patterns of network traffic, especially any odd or questionable behaviour, might help spot possible attempts at data exfiltration.
- User activity logs: Logs of user activity Examining login events, user access records, and file access/modification activities to spot any unusual or suspicious activity.

- **Endpoint devices:** Monitoring endpoints, such as employees' laptops, desktop computers, servers, and increasingly USB drives, is essential for identifying unapproved data transfers and unearthing strategies for data hiding.
- **Cloud collaboration and storage tools:** Examining how cloud-based (Google Drive, OneDrive) collaboration and storage are used to find instances of unapproved data transfer or sharing.
- **Systems for recording and monitoring:** Extensive systems for recording and keeping track of pertinent security occurrences and actions.
- **Email correspondence:** Examine all correspondence, both internal and external, to find any instances of unapproved or suspect data exchange.
- **Physical access logs:** Reviewing physical access logs to find any unauthorised or anomalous access to systems or sensitive areas

2.1.2. Collaboration and Communication with IT and Security Teams:

Investigators work in coordination with IT and security teams to take advantage of their knowledge of traversing the organization's architecture, locating pertinent data sources, and guaranteeing adherence to security procedures. Working closely with these teams gives investigators important insights, and access to necessary resources, and guarantees that technical findings are in line with commercial goals.

Table 1: The below tables show IT and Security Teams' Cooperation and Operational Support During an Investigation

Aspect	Description
Leveraging Insider Knowledge	The IT and security personnel at Dark Clarke are well-versed in the systems, infrastructure, and security guidelines of the business. Working together with them facilitates the swift identification of pertinent data sources and possible evidence sources.
Identifying Anomalies and Deviations	The teams are accustomed to typical user behaviour patterns, system activities, and network traffic patterns. Identifying abnormalities or deviations that can point to malicious activity or illegal access is much easier with regular communication.
Facilitating Access and Data Collection	The teams are essential in granting users access to logs, data sources, and systems that are required. Their support guarantees compliance, data integrity, and acquisition

	that runs smoothly.
Addressing Operational Concerns	Investigators work closely together to design and carry out tasks that minimize disruptions to Dark Clarke's continuing business operations, guaranteeing little impact on user productivity and vital systems.
Implementing Remediation Measures	The teams' knowledge is extremely helpful in converting inquiry findings into concrete actions, such as bolstering access controls, improving monitoring capabilities, or correcting vulnerabilities.

2.1.3. Prioritising data sources:

It can get proof that may have leaked from the firm's data by doing this. Therefore, prioritising data sources is based on several factors.

- **Relevance and Likelihood of Containing Evidence:**
It concentrates on data sources that are most likely to have leaked company information by evaluating the relevance and likelihood of containing evidence. With this focused strategy, it has a better chance of finding and obtaining relevant evidence.
- **Breach Specifics:**
It can prioritize the data sources that are most likely to have been impacted by the breach by considering the specifics of the incident. This guarantees that we concentrate on regions where the likelihood of the leak is highest, resulting in a prompt identification of pertinent evidence.
- **Risk of Spoliation:**
It can identify data sources that are more vulnerable to manipulation by ranking the danger of data spoliation. It can stop more evidence loss or manipulation by protecting these sources first and maintaining the integrity of the evidence for future research.
- **Impact on Operations:**
It will weigh our investigative efforts against the necessity of preserving company continuity by taking the impact on operations into account. It will make sure that our inquiry moves forward smoothly and without undue disturbance by giving priority to data sources that have the least possible impact on operations.

It will improve the likelihood of finding evidence connected to the data leak by ranking data sources in this way, which makes the investigation process go more smoothly.

2.2. Preparation phase:

2.2.1. Establishing Chain of Custody and Evidence Handling:

Establishing a clear chain of custody and putting appropriate evidence-handling procedures in place from the beginning of the investigation are essential to maintaining the integrity and admissibility of the gathered evidence. This includes:

- Recording the gathering and transfer of evidence, together with specifics like the day, time, place, and people involved.
- Putting strong access restrictions and safe storage practices in place to stop evidence from being tampered with or accessed by unauthorized parties.
- Producing forensically sound duplicates ensures data integrity by producing precise copies of the original evidence utilizing forensic imaging software, write-blockers, and other trustworthy techniques.
- Following industry best practices and industry-standard evidence handling rules to guarantee the evidence's admissibility in court or regulatory procedures.

2.3. Acquisition phase:

2.3.1. Data Acquisition Methods:

The procedure for collecting data from several sources is a crucial stage in the evidence collection process of a digital forensics investigation. The techniques for obtaining data from various devices and systems are described in this section.

Digital evidence must be acquired using a variety of tools and acquisition techniques that are specific to the systems in concern. There is no such thing as a one-size-fits-all solution because every system has its own distinct architecture and file systems. For example, specific programs like EnCase, Disk Utility, dd command, and Cellebrite are needed for Windows, macOS, Linux, and mobile devices, respectively. Depending on the features of the system under investigation, several acquisition techniques, including disk imaging, logical acquisition, or memory analysis, must be used. This maintains the integrity of the evidence and guarantees comprehensive data collection. (NIJ, n.d.) (cybersecurity exchange , n.d.)

Table 2: The table below shows the System-specific Digital Forensics Tools and Acquisition Techniques

System	Acquisition method	Tools	Note
Windows	Disk imaging	Encase, FTK imager,	Creates accurate duplicates of drives or partitions.

	Live system acquisition	Volatility, Sysinternals	Gets irregular data from an operational system
macOS	Forensic disk imaging Collecting system logs and artifacts	Disk Utility Macquision, OSXcollector, Sysdiagnose	Creates forensic disk images for macOS. Collects artifacts and system logs.
Linux	Disk imaging Memory analysis	dd command, guymager volatility, Lime	Creates disk images in Linux Examines memories for forensic reasons
Network	Network traffic capture Log files analysis	Wireshark, Tcpdump ELK stack, Splunk	Captures and examines network activity Uses Elasticsearch, Logstash, and Kibana to analyse logs.
Mobile devices	Mobile device acquisition Logical acquisition	Cellebrite, XRY Oxygen forensic detective UFED physical analyser	Gathers information from mobile devices Takes logical information out of mobile devices.
Cloud	Cloud data acquisition API data collection	Amazon S3, Google Cloud Storage, Microsoft Azure Custom scripts or tools	Obtains information stored on cloud services Collects information via cloud APIs

2.3.2. Considerations for Encrypted Data and Cloud Acquisition

Given the growing prevalence of encryption and cloud services in contemporary business operations, considering encrypted data and cloud acquisition is an essential part of the Dark Clarke Company inquiry. The investigation seeks to protect privacy and legal bounds while guaranteeing access to pertinent information using encrypted data. Meanwhile, knowing cloud acquisition is essential since it enables investigators to obtain evidence that is off-site

and frequently spreads over multiple jurisdictions, necessitating adherence to access controls and data security rules. The aforementioned factors facilitate the navigation of intricate digital settings by Dark Clarke investigators, all the while maintaining evidence integrity and legal compliance.

Table 3: the table shows Aspects to Consider When Acquiring Cloud and Encrypted Data

Considerations	Encrypted data acquisition	Cloud acquisition
Key management	Make sure that the passwords or encryption keys used in business systems are accessible. Document the main management procedure.	Recognizes the cloud services' encryption key management practices. Make sure you have access to cloud service providers' encryption keys.
Data recovery tools	Utilizes specialist forensic instruments for deciphering Checks for compatibility with encryption schemes and file systems.	Makes use of the cloud's encryption key recovery techniques. Make sure the tools are compatible with cloud environments.
Legal compliance	Verify adherence to relevant privacy and data encryption laws and regulations. Keep a record of every action you take to obtain and manage encrypted data.	Respect the rules and laws when it comes to getting proof from cloud services. Records the legal procedure that was used to acquire cloud data.
Access controls	Strict access controls should be put in place to stop illegal access to encrypted data. Tracks and records each attempt at access.	Make sure the cloud environment's access controls are set up correctly. Tracks and examines cloud data access.
Data security	Make sure encrypted data is safe to acquire and analyse. For the transmission of data, use secure channels.	Make sure the information kept on the cloud is safe. For both data in transit and data at rest, use encryption.
Jurisdictional issues	Recognize the potential jurisdictional problems while working with encrypted data. Establishes the legal	Takes jurisdictional concerns into account while gaining access to cloud-based data. Observe the applicable

	parameters for data collection.	jurisdictions' legislation.
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2.4. Preservation Phase:

To ensure the confidentiality and integrity of the evidence gathered, a digital forensic investigation's preservation phase is essential. To stop manipulation or unwanted access entails making forensic photographs of digital storage devices and making sure that access control and secure storage are in place. The integrity of the evidence is maintained by following strict preservation methods, which paves the way for in-depth inspection and analysis in later stages. (IBM, n.d.) (exterro, n.d.)

2.4.1. Creating Working Copies and Forensic Images:

- An essential first step in protecting digital material is forensic imaging. It entails making a precise duplicate of the original storage device, often known as a forensic image.
- By guaranteeing that the original evidence is not altered, forensic imaging maintains its integrity for analysis.
- These forensic photos are converted into working copies for review and analysis, reducing the possibility of tampering with the original evidence.
- Forensic photos and functioning copies are produced using programs like EnCase, FTK Imager, and the dd command.

2.4.2. Secure Storage and Management of Access:

- After forensic pictures and functional copies are produced, they need to be kept safely to avoid loss, alteration, or unwanted access.
- The process of storing evidence securely entails keeping it in a monitored space with limited physical and digital access.
- To guarantee that the evidence is handled only by authorized persons, access control procedures are put in place. (NIJ, n.d.)
- To restrict who has access to the evidence and what they can do role-based access control is put in place.
- Techniques for encryption can be used to safeguard the evidence's privacy both during transmission and storage.

- Logging systems are implemented to monitor who has access to the evidence and uphold an unambiguous chain of custody.
- Using hashing algorithms like MD5 or SHA-256, routine integrity checks are carried out to confirm the integrity of the evidence that has been saved. (NIJ, n.d.)
- Strategies for redundancy and backup are used to prevent data loss and guarantee access. (exterro, n.d.)

It will guarantee that the digital evidence is secure, undamaged, and admissible for the duration of the inquiry by following these procedures. This establishes the framework for in-depth investigation and analysis, allowing us to make precise judgments and provide convincing results.

2.5. Adherence to ACPO principles:

The digital forensic examination carried out for Dark Clarke was carried out by the widely accepted ACPO (Association of Chief Police Officers) principles. These principles offer a structure for guaranteeing the accuracy, dependability, and admissibility of digital evidence. The highest standards of forensic procedure will be upheld throughout the investigation process by adhering to these principles. (ACPO, n.d.)

1. Principle of Data Integrity:

- All procedures for gathering, managing, and analysing data will be carried out in a way that protects the original evidence's integrity.
- Write-blocking strategies will be used to avoid accidental alteration or contamination when obtaining data from digital sources.
- To ensure the validity and integrity of the evidence, precise bit-for-bit replicas of the original data sources were created using forensic imaging and hashing.

2. Principle of Audit Trail:

- Throughout the investigation, a thorough audit trail was kept, recording all actions conducted, the instruments and methods used, and the people involved.
- Every activity's date, time, and kind were documented in in-depth logs, along with the individuals in charge of carrying it out. Transparency, accountability, and the capacity to recreate the sequence of events in the event of a necessity are guaranteed by this audit trail. (ACPO, n.d.)

3. The Specialized Expertise Principle:

- Experts in several fields, including digital forensics, comprised the investigation team. They will be all extremely skilled and experienced.

- Outside consultants and subject matter experts will be called upon when specific knowledge or sophisticated analytical skills are required.
- The team maintained current knowledge of the newest forensic procedures and industry best practices thanks to ongoing training and professional development opportunities. (ACPO, n.d.)

4. Safe Storage and Handling Principle:

- All digital evidence collected for the inquiry was handled and stored safely in a controlled setting, with stringent protocols for chain of custody and access control in place.
- The evidence was moved and transferred securely, with the proper paperwork and tamper-evident seals.
- The evidence was physically and logically restricted to authorized personnel alone, protecting its integrity and admissibility. (ACPO, n.d.)

2.6. Operational trade-offs:

The strategic balancing of conflicting aspects including cost, time, and resource allocation is known as an operational trade-off. These trade-offs are critical in decision-making to maximize efficiency and accomplish cybersecurity goals. This entails effectively managing risks, responding to evolving threats, and upholding stakeholder confidence.

The following are some more detailed suggestions and methods for reducing the noted operational trade-offs

a). Cost:

- Set important data sources and areas in order of importance.
- Make use of internal expertise and resources.
- Examine less expensive choices such as open-source or cloud-based forensic software.

b). Time:

- Create a thorough project plan with precise deadlines and completion dates.
- Provide specialized personnel and resources.
- For large data sets, use parallel processing and analysis techniques.

c). Disruption of business operations

- Create a communication plan to notify the appropriate stakeholders.
- Reduce disruptions by planning activities for off-peak hours.
- Put live forensics techniques and virtualization to use.
- Form a dedicated staff to look after important procedures.

d). Legal and regulatory ramifications

- Seek early legal counsel to guarantee adherence.
- Observe chain of custody and evidence handling procedures.
- Create a thorough plan for reporting and documenting.

e). Employee privacy and morale

- Clearly state the bounds and limitations.
- Address problems by communicating regularly.
- Put in place a procedure for anonymous reporting.
- Aid the impacted staff members.

3. Examination and analysis:

Understanding the type and scope of a data breach requires a thorough understanding of the Examination & Analysis phase of a digital forensic investigation. To glean important details about the incident, a thorough inspection and analysis of digital data are required. To accomplish these goals, Dark Clarke's forensic team will work with its IT and security departments to conduct several procedural actions and forensic studies.

3.1. Collection and Preservation of Evidence:

Gathering and storing digital information related to the data breach incident is the first stage in the Examination & Analysis phase. To secure the integrity and acceptability of the data, this entails gathering and safeguarding information from multiple sources. The instruments and methods used are as follows: (exterro, n.d.) (Medium, n.d.) (NIJ, n.d.)

Forensic Imaging: Making forensically sound copies of storage devices with programs like EnCase Forensic and Access Data FTK Imager. This procedure guarantees that the original data is preserved and is admissible as proof in court.

Memory Acquisition: Gathering memory dumps from endpoint devices using the Volatility Framework. Memory analysis offers information on ongoing operations, network connections, and ephemeral material that could be vital to the inquiry.

Analysis of Network Traffic: Network traffic can be captured and examined with Wireshark. This aids in spotting shady connections or trends that point to illegal access or data exfiltration.

To keep evidence intact and admissible for future review and possible legal procedures is the

basic justification for gathering and preserving it. Dark Clarke offers forensically sound copies of digital media and memory to guarantee that evidence is preserved and unchangeable.

3.2. Preparing Forensic Tools and Equipment:

This table lists how the digital forensic tools Volatility Framework, EnCase Forensic, and Wireshark will be used in the investigation to look at memory dumps, analyse endpoints, and find evidence of attempted data exfiltration, respectively. It describes the functions and methods of analysis of each tool concerning the data breach inquiry.

Table 4: Analysis of key digital forensics tools

Tool	Description	Analysis
Wireshark	Identifying attempts at data exfiltration	Records and examines network activity while the data breach is happening. Filters traffic using popular protocols for data exfiltration (HTTP, FTP, SMTP) Spots odd traffic patterns, bulk data transfers, and connections to rogue websites. Examines packet contents and rebuilds data streams to find transmissions of sensitive data. Links suspicious traffic to file access records and user activity logs.
Encase forensics	Endpoint analysis	Obtains forensic pictures of important endpoints (laptops, workstations)

		Checks photos for malware infestations, unauthorized access, and data-hiding strategies. Look for terms, file kinds, and trends associated with data breaches. Reassembles the chronology of incidents and the criminals' acts. Recognize attempts to hide evidence or cover up tracks
Volatility framework	Memory analysis	Obtains and examines memory dumps from important workstations and servers. Retrieves details about active processes, open connections, and loaded DLLs. Locates data staging locations, inserts code, malicious processes, and encryption keys. Finds proof of specialized malware or complex data-hiding techniques

3.3. File system analysis:

3.3.1. Analysing Metadata and File System Structures:

Studying how data is arranged within the file system might help identify any anomalies in the file structures or instances of unwanted access. Unauthorized changes to important system files or strange file creation timestamps, for example, may point to malicious activity.

- This includes understanding the file system's data organization and examining structures such as NTFS's Master File Table (MFT) and UNIX-based systems' inodes.

- Important details including file ownership, permissions, timestamps, and file size are provided by metadata.
- File system structures and metadata can be analysed using programs like Autopsy and The Sleuth Kit (TSK).

3.3.2. Recovering Back Deleted Folders and Files:

Deleted files can have proof of the data leak or disclose the offender(s)'s activities. Finding deleted files might reveal information about what data was viewed, altered, or stolen.

- Frequently, deleted files are not permanently erased from the file system; instead, the space they occupy is designated as free for future usage.
- Autopsy and The Sleuth Kit (TSK) are two tools that can recover deleted files by separating the contents and examining file system metadata.

3.3.3. Finding and Examining Files That Are Hidden or Encrypted:

To hide their actions, offenders may employ complex data-concealing or encryption techniques. Finding and examining encrypted or hidden files is essential to locating proof of the data leak.

- File system analysis can identify hidden files, even if they might not be apparent through typical file browsing.
- Sometimes the existence of encrypted files can be inferred from file metadata or odd access patterns. Encrypted files require decryption keys or passwords for access.
- EnCase and X-Ways Forensics are well-known for their ability to analyse encrypted or concealed files.

Key findings:

The team discovered indications of sensitive files being viewed and copied to external USB drives in the weeks preceding the intrusion after looking through the file system metadata and retrieving erased files from the work machines of important suspects. These machines had questionable after-hours activity, according to timestamp analysis. Large amounts of private customer data, such as contracts, financial records, and proprietary technological designs, were accessed outside of regular business hours, according to forensic examination of the file systems. Additionally, the metadata suggested a staged data exfiltration procedure since it

showed that these files were copied locally to the suspects' workstations before being moved to the external USB devices.

Instruments and Methods

A set of command-line tools for file system investigation is called the Sleuth Kit (TSK).

Autopsy: A graphical user interface for TSK that facilitates analysis.

X-Ways and EnCase Commercial tools with extensive file system investigation capabilities are known as forensics.

3.4. Advanced-Data Hiding Techniques

3.4.1. Detecting Data Hiding and Steganography Techniques:

Investigators believe that one or more workers are stealing sensitive data by employing advanced data-hiding strategies. Finding hidden data and determining the scope of the data leakage depends on the detection of steganography and other data-hiding methods.

- This entails utilizing steganography or other data-hiding techniques to locate and extract hidden data that has been buried inside digital files.
- Steganography conceals data from files that appear harmless, such as audio or picture files, without changing their look or capabilities.
- Visual inspection, statistical analysis, and the use of steganalysis tools to uncover buried data are examples of detection strategies.

3.4.2. Attempting Decryption of Encrypted Files:

During the inquiry, encrypted files will be discovered that might have important information about the data breach. Decryption attempts are necessary to obtain this data and assess its applicability to the inquiry. Investigators must investigate several decryption techniques to unearth concealed information and compile proof against it.

- To access the contents of encrypted files, this procedure focuses on decrypting them.
- Cryptographic algorithms are used to safeguard encrypted files, which cannot be accessed without decryption keys or passwords.
- Brute-force assaults, dictionary attacks, and taking advantage of known vulnerabilities in encryption algorithms are some of the methods used in decoding attempts to recover the decryption key.

3.4.3. Documenting Identification and Analysis Processes:

Transparency and accountability in the investigation are ensured by accurate recording of the identification and analysis processes. It also offers a thorough record of the methods employed and the conclusions drawn, which helps in decision-making and possible legal actions.

- Documenting the identification and analysis procedures used to find concealed or encrypted data is part of this stage.
- Documentation offers specifics about the methods and resources employed, the conclusions, and any difficulties that arose while doing the study.
- Accurate documentation promotes knowledge exchange among forensic analysts and guarantees the transparency and integrity of the investigation.

Key findings:

The use of steganography to conceal data within seemingly innocuous image and audio files was discovered through forensic analysis. Additionally, encrypted archive files were discovered, indicating that the suspects concealed their operations with robust encryption.

3.5. Application and Artifact Analysis

3.5.1. Examining User-Created Files and Documents:

Examining user-created files' metadata, content, and file structures are all part of the analysis process. Sensitive information, signs of unauthorized access, communication logs, and exfiltration efforts will all be searched for during content analysis. Timestamps, author details, and geolocation data will all be carefully examined for irregularities during metadata analysis. The analysis of file structures will search for methods of hiding data, altered or erased files, and indications of manipulation.

Tools and methods specific to the analysis of dubious content—like encrypted files or buried data—will be used. File structure anomalies, such as odd headers or footers, will be examined for signs of manipulation or obfuscation attempts.

The results will be cross-checked against logs, network traffic, and memory dumps, among other sources of evidence. A thorough chronology that includes information on activities, access patterns, and possible perpetrator acts will be recreated. Finding data breaches,

locating unauthorized activity, identifying the culprits of the data breach, and comprehending the incident's methodology and extent are the objectives of this analysis.

In addition to supporting the identification of evidence and accountable parties, the investigation of content, metadata, and file structure, in conjunction with correlation analysis and timeline reconstruction, will offer a comprehensive picture of the data breach.

3.5.2. Analysing Application-Specific Artifacts:

This stage entails looking into artifacts—like logs, cache files, and configuration files—that have been left behind by certain programs. These artifacts frequently hold important details like user behaviour, application usage trends, and possible security holes. Investigators can learn more about how programs will be used, what was done, and whether any suspicious activity took place by examining these artifacts. This analysis aids in comprehending the actions of both authorized users and possible intruders. (Gostojić, 2023)

Purpose: Employees may have used a variety of apps, some of which may have left behind artifacts with important data. By examining these artifacts, it can gain knowledge about user behaviour, program usage trends, and possible security holes. Investigators can examine chat logs, email client data, online browser history, and other application-specific information to find oddities or trends associated with the data leak.

3.5.3. Reconstructing User Activity Timelines:

Using information from multiple sources, a chronological timeline of occurrences is assembled to create user activity timelines. This comprises application logs, network logs, file access logs, and other pertinent data sources. Investigators can create a comprehensive timeline of user activities, including file accesses, network connections, and login sessions, by cross-referencing this data. This timeline helps identify any illegal activity, strange patterns, or potential security breaches by giving a thorough summary of user interactions with the system.

Purpose: To comprehend the series of events leading up to the data leak, Dark Clarke reconstructs user activity timelines. It enables us to create a timeline of user activities, such as network activity, file access, and application usage. This makes it easier to pinpoint the offender(s) and their methods of operation. (CASINO, 2021)

Key findings:

The team was able to put together user activity timelines by assembling forensic artifacts such as browser history, email/chat logs, and application usage logs. These demonstrated a pattern of the suspects deleting logs, speaking with unidentified third parties, and gaining access to private information when not doing their regular tasks.

3.6. Network and cloud analysis:

3.6.1. Analysing configuration files and network logs:

- Configuration files, which hold parameters that dictate how devices operate, are crucial for managing network activity. Firewall rules, for example, control what traffic is allowed and prohibited, preventing illegal access by blocking ports or IP addresses. In a similar vein, routing tables establish traffic paths that devices take to get to different locations on the network.
- A thorough history of network traffic is provided by network logs. They contain the protocols that will be used (e.g., TCP, UDP), source and destination IP addresses, and timestamps showing the times of the events. These logs help with security analysis, monitoring, and troubleshooting by offering insightful information about network behaviour.
- Finding typical problems requires analysing network logs and configuration data. This entails locating instances of unwanted access, such as persistently unsuccessful login attempts or dubious IP addresses in firewall logs. Network logs also show strange traffic patterns, such as sudden activity spikes or abnormally massive data transfers. Analysis can also reveal configuration flaws that could jeopardize network stability or security, such as incorrectly configured firewall rules or routing issues.

To find any unusual network activity or misconfigurations, investigators may carefully examine configuration files and network logs. This involves keeping an eye out for any attempts at illegal access, odd traffic patterns, or configuration mistakes that might result in security flaws.

3.6.2. Analysing Recorded Network Activity:

- To find trends, abnormalities, and possible security risks, this procedure comprises recording and examining network traffic.
- Data packets sent and received between networked devices are included in captured network traffic.
- Protocol analysis, anomaly detection, and traffic pattern recognition are examples of analysis approaches.

It should keep an eye out for indications of malware infestations, data exfiltration, and other security breaches by recording and examining network traffic. It can identify malicious activity and minimize threats by studying network patterns and packet contents.

Key findings:

Large encrypted file transfers to an external IP address occurred on several occasions within the same period, according to an analysis of intercepted network data. These transfers got

across firewalls and other standard security measures. The encrypted nature implied the employment of data concealment strategies.

3.6.3. Examining Data from Remote Servers and Cloud Storage:

- This entails looking for signs of illegal access or data espionage in data kept on remote servers and cloud platforms.
- Configuration settings, logs, databases, and files can all be found in cloud storage and remote server data.
- Examining access logs, examining file information, and looking for odd account behaviour are some methods of investigation.

It will investigate data from external servers and cloud storage to see if any private information has been hacked or released. It can detect illegal access, data leaks, or security breaches coming from cloud platforms or remote servers by looking over access logs and examining stored data. This aids in protecting private data and averting occurrences in the future.

Key findings:

Analysing cloud storage records associated with the same primary suspects showed that, in the days preceding the breach's discovery, they had exchanged and synchronized sensitive data to personal cloud accounts such as google drive.

In summary, cyber security strategy relies heavily on network and cloud analysis, which enables keeping an eye on network activity, identifying potential security risks, and safeguarding critical data kept on distant servers and cloud platforms.

3.7. Mitigation:

- To reduce the potential for insider attacks and data exfiltration, Dark Clarke needs to put in place a thorough security plan. This entails routinely assessing user access rights and bolstering access controls with multi-factor authentication and role-based access.
- Programs for employee awareness and training are crucial for educating staff members about social engineering and cybersecurity threats.
- To identify and address suspicious activity, monitoring, and detection technologies like intrusion detection, SIEM, and user behaviour analytics should be implemented.
- Strong encryption techniques should be used to encrypt sensitive data while it is in transit and at rest.
- To guarantee readiness, Dark Clarke should also regularly practice drills and exercises. Lastly, a strong incident response plan should be established, detailing protocols for event detection, containment, and recovery.

4. Conclusion:

4.1. Reporting and presenting phase:

In summary, the results of this study show that for DarkClarke to properly handle the suspected insider threat and data exfiltration incident, a thorough digital forensic investigation is necessary. Through the utilization of industry best practices and a methodical approach, DarkClarke can obtain crucial evidence to identify the origin of the data breach and put preventative measures in place to avoid it happening again.

Key phases of the inquiry are covered, such as data reduction and preprocessing, file system analysis, application and artifact analysis, forensic acquisition, examination, and analysis, and network and cloud analysis. Every step is covered in detail, with a focus on the significance and applicability of different methods, resources, and factors.

Ensuring the integrity of the evidence and following legal and ethical requirements are critical to the success of the inquiry. It is important to establish a correct chain of custody, use forensically sound techniques, and carefully document all procedures and conclusions.

DarkClarke can detect questionable network and cloud behaviour, unearth complex data-hiding tactics, and recreate user activity timelines through comprehensive research. These results will be useful in identifying the offender or offenders as well as informing future efforts to improve the cybersecurity posture of the organization.

The investigation prioritizes gathering solid evidence with cutting-edge forensics technologies to meet the obstacles presented by highly competent threat actors. Methods such as data carving and memory forensics are applied, with an emphasis on locating advanced anti-forensic techniques.

The investigation's wider ramifications are also considered in the report, including possible effects on costs, business continuity, privacy, and admissibility of evidence. Suggestions for continuous security monitoring and improvement are included, along with recommendations for phased resource allocation, operational impact mitigation, and stringent evidence management protocols.

To summarise, the research findings emphasise the significance of conducting a comprehensive digital forensic examination when handling cybersecurity events. A detailed action plan for locating attack routes, addressing vulnerabilities, and strengthening

DarkClarke's cybersecurity posture is given. The well-reasoned conclusion shows a deep comprehension of the gravity of the issue and the required response.

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